Pre-Characterization Process Risk Assessment

A-Mab Drug Substance — RA-001

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
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Abbreviations

ADCC, antibody-dependent cell-mediated cytotoxicity; AEX, anion exchange chromatography; AMV, analytical method validation; BLA, Biologics License Application; CEX, cation exchange chromatography; CPP, critical process parameter; CQA, critical quality attribute; DNA, deoxyribonucleic acid; DoE, design of experiments; DS, drug substance; FMEA, failure mode and effects analysis; GPP, general process parameter; HCP, host cell protein; HMW, high molecular weight; ICH, International Council for Harmonisation; icIEF, imaged capillary isoelectric focusing; IgG1, immunoglobulin G subclass 1; KPP, key process parameter; LRV, log reduction value; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; PAR, proven acceptable range; PCMP, Process Characterization Master Plan; PCMR, Process Characterization Master Report; PCP, Process Characterization Plan; PCR, Process Characterization Report; pCO2, dissolved carbon dioxide partial pressure; PPQ, process performance qualification; PTP, Process Transfer Plan; QbD, quality by design; RA, risk assessment; RPN, risk priority number; RRF, risk ranking and filtering; SEC, size-exclusion chromatography; SOP, standard operating procedure; TMP, transmembrane pressure; UF/DF, ultrafiltration and diafiltration; WC-CPP, well-controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

1 Purpose and scope

This assessment decides which A-Mab process parameters are characterized, and by what kind of study. It is performed before any characterization study is executed, and its output is the study scope that the Process Characterization Plans then carry out.

The A-Mab drug-substance process is being transferred from the development site to the commercial drug-substance site under PTP-001. Characterization at commercial-equivalent scale-down is the evidence that supports that transfer and the operating ranges filed with it, and because characterization effort is finite, the purpose of this assessment is to place that effort where the risk to product quality lies and to record why the remaining parameters need less.

The assessment covers the 37 process parameters of the drug-substance train (Steps 3 to 10). For each parameter it records a potential failure mode, the quality attributes that failure mode could reach, a pre-characterization risk score and an assigned study type. Raw-material attributes, equipment qualification and analytical method performance are outside its scope and are addressed by PTP-001 and by the method validation records (AMV series). The drug product process is outside the scope of the whole characterization package.

Two things this document does not do should be clear at the outset. It does not classify any parameter, because the critical, well-controlled critical, key and general process parameter designations are outputs of the characterization studies, and they are assigned in the individual reports for each step (PCR-003 through PCR-010) and consolidated in PCMR-001. It also calculates no residual risk, since the control strategy that would reduce the risk is not yet defined. What is recorded here is the risk as it stands, with the controls that exist today.

The documents that frame this one are listed in Table 1.1.

Table 1.1: Parent and downstream documents of this assessment.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

2 Risk assessment methodology

The method ranks each parameter twice and uses the combined score to choose a study type. The first ranking is the potential impact of the parameter on a quality attribute, and the second is the potential for that parameter to interact with other parameters of the same unit operation. This is the risk ranking and filtering approach of the A-Mab case study (CMC Biotech Working Group 2009), applied at the point in the lifecycle (Stage 1, process design) at which the commercial process is established (U.S. Food and Drug Administration 2011).

Two assessments are kept separate throughout, because they answer different questions. The criticality of a quality attribute is a property of the molecule and of the patient, and it is assessed on impact and uncertainty alone (Tool #1 of the case study). The risk carried by a process parameter is a property of the process, and it uses severity, occurrence and detection. Confusing the two would let manufacturing capability change the criticality of an attribute, and criticality must not depend on how well a particular facility happens to control a parameter.

Criticality is treated as a continuum and not as a binary label (International Council for Harmonisation 2023b). Every attribute carries a criticality level, and the attributes at high and very high criticality are treated as critical for the purposes of this assessment (CMC Biotech Working Group 2009). The consequence for a process parameter is that its severity is inherited from the most critical attribute it can affect, so a parameter that reaches several attributes takes the severity of the worst of them and a parameter linked to no attribute at all sits at the bottom of the scale.

The three scoring scales are given in Table 2.1. Severity describes the consequence of the failure mode for the product, occurrence describes how often the parameter is expected to leave its range with the controls that exist today, and detection describes whether the resulting change would be seen and where in the process or in the testing programme it would surface.

Table 2.1: Severity, occurrence and detection scales (CMC Biotech Working Group 2009).

Scale	Score	Band	Definition
Severity	10	Very High	Definite impact to product quality; lot(s) rejected
Severity	8	High	Probable impact; significant supplemental testing / re-processing
Severity	6	Moderate	Potential impact; supplemental t=0 testing / minor re-processing

Scale	Score	Band	Definition
Severity	4	Slight	Unlikely impact; no supplemental testing / no re-processing
Severity	2	Low/None	No impact to product quality
Occurrence	10	Very High	Failure ~ once per 100 units
Occurrence	8	High	Failure ~ once per 1,000 units
Occurrence	6	Moderate	Failure ~ once per 2,000 units
Occurrence	4	Low	Failure ~ once per 5,000 units
Occurrence	2	Minimal	Failure ~ once per 10,000 units or lower
Detection	10	None	Not detected by in-process or CofA testing
Detection	8	Low	Not detected in-process; caught by CofA testing
Detection	6	Moderate	Detected downstream before CofA, not during the unit op
Detection	4	High	Detected in the next downstream operation
Detection	2	Very High	Immediately detected by in-process monitoring / inspection

Occurrence and detection are scored as they stand before characterization, and the assignment is given in Table 2.2. A parameter that can affect a quality attribute takes the higher occurrence band, because the range that would keep the attribute inside its limit is not yet known and the current range is not yet proven. Detection is scored from where the failure would surface. A glycan or charge change is not visible during the operation that forms it and is found at release testing (the second detection band). A loss of viral clearance is not visible at all on the drug substance, since clearance is demonstrated in small-scale spiking studies and is never measured on the batch (International Council for Harmonisation 2023a), so the 8 parameters that can reach a viral-clearance attribute take the lowest band.

Table 2.2: Initial occurrence and detection assignment, before characterization.

Parameter group	Occurrence (initial)	Detection (initial)	Parameters
Viral-clearance attribute at risk	7	10	8
Product-quality attribute at risk	7	8	13
Process performance only	4	6	16

The product of the three scores is the initial risk priority number, which ranks parameters against one another and is not a probability. Priority follows from severity. A parameter that can affect an attribute at the high severity band or above is ranked High priority, a parameter linked only to a less severe attribute is ranked Medium, and a parameter that acts on process performance alone is ranked Low.

The study type is then decided by the combined risk score. Both rankings use the impact scale in Table 2.3, and their product is mapped to a study type by the bands in Table 2.4. Parameters at the top band are studied together in a multivariate design, because an effect that depends on the level of a second parameter cannot be estimated one factor at a time. Parameters in the middle band go to a multivariate design as well, unless a univariate study can be justified from the mechanism by which the parameter acts and from the controls already placed on it. Parameters at the bottom of the scale need no new study, and their ranges are supported by platform knowledge and by the existing manufacturing controls (procedural and equipment).

Table 2.3: Impact scale used for both rankings (CMC Biotech Working Group 2009).

Ranking	Score	Basis
No Impact	2	Variation not detectable/within assay variability
Minor Impact	4	Within acceptable range
Major Impact	8	Outside acceptable range, can be near edge of failure

Table 2.4: Combined risk score and the study type it assigns (CMC Biotech Working Group 2009).

Combined risk score	Assigned study type
32 and above	Multivariate
8-16	Multivariate or justified univariate
4	Univariate
2 and below	No study

Where no data or rationale was available to score a parameter, the parameter was ranked at the highest level (CMC Biotech Working Group 2009). This conservative default gives the method a deliberate asymmetry, since uncertainty can move a parameter up the scale but cannot move it down, and the assignment consequently errs toward more study. The cost of that error is study effort. The opposite error would leave a real effect undetected until process performance qualification (PPQ), where the cost is a failed batch and a delayed filing.

3 Quality attributes at risk

The severity of every parameter in this assessment is inherited from the attribute it can affect, so the attribute register is the input to the ranking rather than a summary of it. The register is given in Table 3.1, with the criticality level assigned to each attribute, the score behind that level, the severity it confers on a parameter, and the step that forms or sets it.

Table 3.1: Quality attributes, criticality and the severity each confers.

Quality attribute	Category	Acceptance	Criticality	Tool #1	Severity	Set by
Afucosylation	glycosylation	2–13 % of glycans	H	60	8	Production Bioreactor
Galactosylation (total %Gal)	glycosylation	10–40 % (G1+G2 equiv.)	H	48	8	Production Bioreactor
High Mannose	glycosylation	3–10 % of glycans	VH	80	10	Production Bioreactor
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60	8	Production Bioreactor
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4	4	Production Bioreactor
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36	6	Production Bioreactor
Residual DNA	process impurity	0–0.001 ng/dose	VL	6	2	Production Bioreactor
Leached Protein A	process impurity	0–5 ppm	L-M	16	4	Protein A Chromatography
Viral clearance — XMuLV (enveloped)	viral safety	16.7–30 log10 (cumulative)	VH	20	10	Low-pH Viral Inactivation
Viral clearance — MVM (parvovirus)	viral safety	8.6–20 log10 (cumulative)	VH	20	10	Anion Exchange Chromatography

10 attributes are carried, and 6 of them are at high or very high criticality. 3 sit at very high criticality, namely high mannose and the cumulative viral-clearance attributes

(XMuLV for enveloped virus and MVM for parvovirus), which together set the top of the severity scale for every parameter that reaches them.

The glycan attributes are linked to efficacy. A-Mab is a humanized IgG1 whose principal mechanism of action is antibody-dependent cell-mediated cytotoxicity (ADCC), so afucosylation and galactosylation change potency directly, and high mannose changes clearance in addition to potency. All three are formed in the production bioreactor, and prior platform experience shows that they are not modified by the purification train, because affinity capture under platform conditions does not separate glycoforms and the charge-based steps (cation and anion exchange) do not discriminate between them either (CMC Biotech Working Group 2009). The severity of the whole glycan set falls on one step, which is the single largest influence on the shape of this assessment.

The viral-clearance attributes behave differently again. Both are cumulative over the process and neither can be measured on the drug substance, so they are demonstrated in small-scale spiking studies at the steps credited with clearance under the modular approach, which are the low-pH hold, anion exchange and virus filtration (International Council for Harmonisation 2023a). A parameter at any one of those steps inherits the severity of a very high criticality attribute together with the lowest detection score, and that combination produces the highest initial risk numbers in the assessment before any evidence about the size of the parameter's effect has been gathered.

The process impurities are shared across steps. Host cell protein (HCP) and residual DNA are formed in the culture and are cleared by three chromatography steps in series, so no single step carries either attribute alone, and the ranking reflects that. Host cell protein appears against parameters of the capture step, the cation exchange step and the anion exchange step, and it confers the same severity in each. How the clearance is apportioned between those steps is decided by the studies (PCP-005, PCP-007 and PCP-008) and consolidated in PCMR-001, not here.

Two attributes are retained at very low criticality. Acidic charge variants and residual DNA sit at the bottom of the register, and both are still ranked, because an attribute of low criticality still has an acceptance limit and a parameter that can reach that limit still needs a range. Acidic variants are the reason the culture chemistry parameters are ranked beyond their glycan effects, and residual DNA, together with host cell protein, is the reason culture duration is ranked at all. Leached Protein A sits between the two groups at low to moderate criticality and is set by the capture step, where it is measured by the leached Protein A immunoassay (AMV-3016).

4 Parameter risk ranking by unit operation

37 parameters were carried into the assessment across the 8 steps of the drug-substance train. 21 of them can affect at least one quality attribute, and the remaining 16 act on process performance only. 16 parameters are ranked High priority, 5 Medium and 16 Low. The highest initial risk number in the assessment is 700, and the 8 parameters that reach it all belong to the three steps credited with viral clearance (the low-pH hold, anion exchange and virus filtration).

The tables below give, for each step, the failure mode considered, the attributes that failure mode could reach and the resulting score, and the narrative names the parameters that drive the outcome of each step and the mechanism by which they act. Existing controls are described in the text, since it is the control already in place, together with the monitoring already in place, that fixes the occurrence and detection scores.

4.1 Production bioreactor

The bioreactor forms 7 of the 10 attributes in the register, so it carries more severity than any other step in the train. 5 of its 9 parameters are linked to a quality attribute.

Table 4.1: Pre-characterization risk ranking, production bioreactor.

Parameter	Potential failure mode	Attribute(s) at risk	Sev.	Init. RPN
Culture pH	pH excursion outside the characterized range (6.6-7.1)	Afucosylation, Galactosylation (total %Gal), High Mannose, Acidic charge variants (deamidation), Aggregates (HMW)	10	560
Culture temperature	Temperature excursion outside 34-36 C	Afucosylation, Galactosylation (total %Gal), Acidic charge variants (deamidation)	8	448
Dissolved CO2 (pCO2)	Dissolved CO2 outside 40-160 mmHg	Galactosylation (total %Gal), Acidic charge variants (deamidation)	8	448
Osmolality	Osmolality outside 360-440 mOsm	Galactosylation (total %Gal), Acidic charge variants (deamidation)	8	448

Parameter	Potential failure mode	Attribute(s) at risk	Sev.	Init. RPN
Culture duration	Culture extended beyond 15-19 days / harvested on low viability	Afucosylation, Galactosylation (total %Gal), Aggregates (HMW), Host Cell Protein (HCP), Residual DNA	8	448
Dissolved oxygen	DO outside 30-70%	process performance	4	96
Initial viable cell conc.	Inoculation density outside 0.5-1.5e6/mL	process performance	4	96
Basal medium concentration	Basal medium concentration outside 0.8-1.6X	process performance	4	96
Nutrient feed-1 volume	Nutrient feed-1 volume outside 9-15% WV	process performance	4	96

Culture pH is the only parameter that reaches all five formed attributes, and it takes the highest severity of the step. Temperature acts with pH on the glycosyltransferase activity that sets afucosylation and galactosylation, and the two are expected to interact. Dissolved carbon dioxide (pCO_2) and osmolality act on galactosylation and on the acidic charge variants, and both change the ionic and buffering environment that pH also changes, so an interaction is expected there as well. Culture duration is different in kind. It does not change the culture chemistry directly but sets how long the product is exposed to that chemistry, and as viability falls toward the end of the run the culture releases host cell protein and DNA while the aggregate level rises.

The remaining parameters act on titer and on run-to-run consistency. Dissolved oxygen, inoculation density, basal medium concentration and nutrient feed volume are held by closed-loop control or delivered gravimetrically, and none of them is expected to move a quality attribute across the range in which it is held. They are ranked at the performance severity band and assigned univariate assessment.

Detection is the weakest element at this step, and parameter control is not. pH, temperature and dissolved oxygen are held against calibrated redundant probes with continuous in-process monitoring, and nutrient feed is delivered against a gravimetric target. None of the five formed attributes is measured during the culture, however, and all of them are seen first at release testing, so the occurrence and detection scores reflect the monitoring of the attribute and not the control of the parameter. 5 parameters go to the multivariate design and 4 to univariate assessment, and both sets are executed under PCP-003.

4.2 Harvest and clarification

No parameter of the harvest step can change a product quality attribute, so every parameter is ranked at the performance severity band and the step takes the lowest initial risk number in the assessment.

Table 4.2: Pre-characterization risk ranking, harvest and clarification.

Parameter	Potential failure mode	Attribute(s) at risk	Sev.	Init. RPN
Centrifugation (rcf)	Centrifugation rcf outside 6000-12000 g	process performance	4	96
Depth filter load	Depth filter load outside 80-160 L/m2	process performance	4	96
Post-clarification turbidity	Post-clarification turbidity high	process performance	4	96

Centrifugation speed and depth filter load act on clarification efficiency, on filter capacity and on step yield. Post-clarification turbidity is the measure of the feed quality handed to capture, and it is monitored in process by nephelometry (AMV-3015). Host cell protein and DNA pass through the step and are cleared downstream, so the risk they carry is ranked against the parameters of the capture and polishing steps and not against these. All 3 parameters are assigned univariate assessment under PCP-004.

4.3 Protein A chromatography

2 of the 6 capture parameters are linked to a quality attribute, and both are linked to the same one.

Table 4.3: Pre-characterization risk ranking, Protein A chromatography.

Parameter	Potential failure mode	Attribute(s) at risk	Sev.	Init. RPN
Protein load	Protein load above characterized range (>50 g/L resin)	Host Cell Protein (HCP)	6	336
Elution buffer pH	Elution pH below characterized range (<3.2)	Host Cell Protein (HCP)	6	336
Load flow rate	Load flow rate high with high load	process performance	4	96
End of pool collect	End-of-pool collection outside 2.0-3.2 CV	process performance	4	96
Operating temperature	Parameter outside characterized range	process performance	4	96

Parameter	Potential failure mode	Attribute(s) at risk	Sev.	Init. RPN
Bed height	Parameter outside characterized range	process performance	4	96

Protein load and elution buffer pH both act on the host cell protein content of the pool, and platform experience indicates that they act together, with a high load at a low elution pH giving the highest pool level (CMC Biotech Working Group 2009). Host cell protein is cleared again at cation and anion exchange, so the severity carried here sits at the moderate band and not at the high band, and the detection score reflects release testing by the host cell protein immunoassay (AMV-3012).

Load flow rate and end-of-pool collection act on yield and on pool volume, and they are ranked at the performance band. Both are nevertheless carried into the multivariate design, because they share a column with the two quality-linked parameters and the composition of the pool cannot be attributed to load alone when the collection window and the residence time move with it. Operating temperature and bed height are held at platform values, are supported by the platform performance history, and are assessed univariately. The step contributes 4 parameters to a multivariate design and 2 to univariate assessment under PCP-005.

4.4 Low-pH viral inactivation

3 of the 4 parameters of the hold step reach the enveloped-virus clearance attribute, and all of them take the maximum initial risk number of the assessment.

Table 4.4: Pre-characterization risk ranking, low-pH viral inactivation.

Parameter	Potential failure mode	Attribute(s) at risk	Sev.	Init. RPN
Inactivation pH	pH > 4.0 (incomplete inactivation) or pH < 3.2 (aggregation/precipitation)	Viral clearance — XMuLV (enveloped), Aggregates (HMW)	10	700
Hold time	Hold time < 60 min (incomplete) or > 180 min (aggregation)	Aggregates (HMW), Viral clearance — XMuLV (enveloped)	10	700
Temperature	Temperature < 15 C	Viral clearance — XMuLV (enveloped)	10	700
A-Mab concentration	Concentration > 35 g/L	process performance	4	96

Inactivation pH has the narrowest acceptable window of any parameter in the process. Above the window inactivation is not assured within the hold time, and below it the product aggregates and may precipitate. Hold time fails in both directions for the same pair of reasons, since a short hold risks incomplete inactivation and a long hold raises aggregate. Temperature acts on the rate of inactivation, so the failure direction is a low temperature at the short end of the hold. A-Mab concentration in the pool is set by the capture step and is bounded by it, and it is ranked at the performance band.

The step is well controlled. The pool is titrated to a target pH under in-line measurement, and the hold is timed with alarms, so occurrence is not what drives the score here. Severity and detection are. Viral clearance cannot be verified on the batch, so the assessment treats every parameter that could reduce it as high priority, as described in §2. That is a deliberately conservative position, and it is the reason a step with 4 parameters and a long platform history (X-Mab, Y-Mab and Z-Mab) nevertheless produces 3 of the highest-ranked parameters in the process. All 3 of them go to the multivariate design under PCP-006.

4.5 Cation exchange chromatography

Cation exchange is the only step in the train that reduces aggregate, so it carries that attribute alone and 4 of its 5 parameters are ranked against a quality attribute.

Table 4.5: Pre-characterization risk ranking, cation exchange chromatography.

Parameter	Potential failure mode	Attribute(s) at risk	Sev.	Init. RPN
Protein load	Protein load above characterized range (>40 g/L resin)	Aggregates (HMW), Host Cell Protein (HCP)	8	448
Load/Wash conductivity	Load/wash conductivity below 3 mS/cm	Host Cell Protein (HCP)	6	336
Elution buffer pH	Elution pH outside 5.8-6.2	Aggregates (HMW)	8	448
Elution stop collect	Extended elution stop-collect (>1.5 OD)	Aggregates (HMW)	8	448
Elution flow rate	Elution flow rate outside 100-300 cm/hr	process performance	4	96

Protein load, elution buffer pH and elution stop collect all reach aggregate and take its severity, and load and wash conductivity reaches host cell protein at the moderate band. Protein load appears against both attributes, which is the clearest interaction expectation at the step, since the column loading that governs how well aggregate is resolved from monomer also governs how much impurity remains bound to the resin when elution begins. Elution pH and stop collect define the same elution peak from two directions, so their effects on the pool are not separable one at a time.

There is no aggregate-reducing step after this one, so the aggregate level of the drug substance is set here. That does not change the severity score, which comes from the attribute and not from the position in the train. It is the reason the interaction ranking is high for every quality-linked parameter of this step, and 4 of them are assigned to one multivariate design. Elution flow rate acts on peak shape and on yield and is assessed univariately (PCP-007).

4.6 Anion exchange chromatography

Every parameter of the anion exchange step is linked to a quality attribute, and 3 of them are linked to both viral-clearance attributes at once.

Table 4.6: Pre-characterization risk ranking, anion exchange chromatography.

Parameter	Potential failure mode	Attribute(s) at risk	Sev.	Init. RPN
Load pH	Load pH below characterized range (<7.2)	Host Cell Protein (HCP), Viral clearance — XMuLV (enveloped), Viral clearance — MVM (parvovirus)	10	700
Equil/Wash-1 conductivity	Equil/Wash-1 conductivity above 3.6 mS/cm	Host Cell Protein (HCP)	6	336
Load conductivity	Load conductivity above characterized range	Viral clearance — XMuLV (enveloped), Viral clearance — MVM (parvovirus)	10	700
Protein load	Protein load above characterized range (>300 g/L resin)	Host Cell Protein (HCP)	6	336
Operating flow rate	Operating flow rate above 450 cm/hr	Viral clearance — XMuLV (enveloped), Viral clearance — MVM (parvovirus)	10	700

Load pH and load conductivity govern the electrostatic environment in which impurities and virus bind to the resin while the antibody flows through, and they are expected to interact, because they change the same environment by different means, and both take the maximum initial risk number of the assessment. Equilibration and wash conductivity and protein load act on host cell protein removal at the moderate severity band, and load also determines how much impurity the resin is asked to hold.

Operating flow rate is the one parameter in the whole assessment that is linked to a quality attribute and is nevertheless assigned a univariate study. Flow acts on virus clearance through residence time and through no other mechanism, and residence time does not interact with the ion-exchange chemistry that the other parameters change. The range is already bounded by the conditions under which viral clearance is validated, so a univariate assessment against that bound is sufficient and a place in the multivariate design would add nothing. The justification is recorded here and carried into PCP-008, where the assessment against the validated bound is executed. The remaining 4 parameters form the multivariate design.

4.7 Small-virus retentive filtration

Both parameters of the virus filtration step are linked to the parvovirus clearance attribute, and both take the maximum initial risk number.

Table 4.7: Pre-characterization risk ranking, small-virus retentive filtration.

Parameter	Potential failure mode	Attribute(s) at risk	Sev.	Init. RPN
Filtration volume (load)	Volumetric load above 105 L/m2	Viral clearance — MVM (parvovirus)	10	700
Filtration pressure	Pressure outside filter operating limits	Viral clearance — MVM (parvovirus), Viral clearance — XMuLV (enveloped)	10	700

Volumetric load is the parameter of record. Removal at this step is by size exclusion, and breakthrough of the smallest model virus (MVM) becomes possible as the membrane accumulates load, so the failure mode considered is an excursion above the validated volumetric limit. Filtration pressure acts on the same membrane and is ranked against both viral attributes. With 2 parameters and one dominant mechanism the multivariate design is small, and it is described in PCP-009. The step is also protected by a post-use filter integrity test, which is a control on the filter rather than on either parameter.

4.8 Ultrafiltration and diafiltration

No parameter of the ultrafiltration and diafiltration step forms or clears a quality attribute, so all 3 are ranked at the performance severity band.

Table 4.8: Pre-characterization risk ranking, ultrafiltration and diafiltration.

Parameter	Potential failure mode	Attribute(s) at risk	Sev.	Init. RPN
Number of diavolumes	Diavolumes < 7	process performance	4	96
Transmembrane pressure	TMP outside 10-15 psi	process performance	4	96

Parameter	Potential failure mode	Attribute(s) at risk	Sev.	Init. RPN
Final DS concentration	Final concentration outside 65-85 g/L	process performance	4	96

The diavolume count sets the completeness of buffer exchange, transmembrane pressure (TMP) sets flux and processing time, and the final concentration is the concentration at which drug substance is delivered to the drug product process, so all 3 parameters are assigned univariate assessment under PCP-010. The formulation characterization itself is reported with the drug product and is outside the scope of this package.

5 Characterization study assignment

22 parameters are assigned to a multivariate design, 14 to univariate assessment and 1 to a justified univariate assessment. 0 parameters are excluded from study. The distribution across the train is given in Table 5.1 and the full assignment in Table 5.2.

Table 5.1: Assigned study types and the steps at which they fall.

Assigned study type	Parameters	Process steps
multivariate DoE	22	3, 5, 6, 7, 8, 9
univariate (justified)	1	8
univariate	14	3, 4, 5, 6, 7, 10

Parameters go into a multivariate design when their effect on an attribute cannot be separated from the effect of another parameter of the same unit operation. That is the case wherever two parameters act on the same attribute through the same physical mechanism, as culture pH and temperature do at the bioreactor and as load pH and load conductivity do at anion exchange (§4.1 and §4.6 respectively). Parameters are assessed univariately when they act on process performance alone, or when a single mechanism accounts for the whole of their effect on an attribute and that mechanism is already bounded. Only one parameter in the assessment falls into the second of those cases, and it is the anion exchange flow rate discussed in §4.6.

The multivariate work falls at 6 of the 8 steps, namely Steps 3, 5, 6, 7, 8, 9, and the 2 steps without a multivariate design (harvest and clarification, and ultrafiltration and diafiltration) have no parameter linked to a quality attribute. 2 parameters are carried into a multivariate design although they are ranked at the performance band, and both are capture parameters carried for the reason given in §4.3.

Table 5.2: Characterization study assignment for every parameter.

Step	Unit operation	Parameter	Sev.	Init. RPN	Prior-ity	Study type
3	Production Bioreactor	Culture pH	10	560	High	multivariate DoE
3	Production Bioreactor	Culture temperature	8	448	High	multivariate DoE
3	Production Bioreactor	Dissolved CO2 (pCO2)	8	448	High	multivariate DoE
3	Production Bioreactor	Osmolality	8	448	High	multivariate DoE

Step	Unit operation	Parameter	Sev.	Init. RPN	Prior-ity	Study type
3	Production Bioreactor	Culture duration	8	448	High	multivariate DoE
3	Production Bioreactor	Dissolved oxygen	4	96	Low	univariate
3	Production Bioreactor	Initial viable cell conc.	4	96	Low	univariate
3	Production Bioreactor	Basal medium concentration	4	96	Low	univariate
3	Production Bioreactor	Nutrient feed-1 volume	4	96	Low	univariate
4	Harvest and Clarification	Centrifugation (rcf)	4	96	Low	univariate
4	Harvest and Clarification	Depth filter load	4	96	Low	univariate
4	Harvest and Clarification	Post-clarification turbidity	4	96	Low	univariate
5	Protein A Chromatography	Protein load	6	336	Medium	multivariate DoE
5	Protein A Chromatography	Elution buffer pH	6	336	Medium	multivariate DoE
5	Protein A Chromatography	Load flow rate	4	96	Low	multivariate DoE
5	Protein A Chromatography	End of pool collect	4	96	Low	multivariate DoE
5	Protein A Chromatography	Operating temperature	4	96	Low	univariate
5	Protein A Chromatography	Bed height	4	96	Low	univariate
6	Low-pH Viral Inactivation	Inactivation pH	10	700	High	multivariate DoE
6	Low-pH Viral Inactivation	Hold time	10	700	High	multivariate DoE
6	Low-pH Viral Inactivation	Temperature	10	700	High	multivariate DoE
6	Low-pH Viral Inactivation	A-Mab concentration	4	96	Low	univariate
7	Cation Exchange Chromatography	Protein load	8	448	High	multivariate DoE
7	Cation Exchange Chromatography	Load/Wash conductivity	6	336	Medium	multivariate DoE
7	Cation Exchange Chromatography	Elution buffer pH	8	448	High	multivariate DoE
7	Cation Exchange Chromatography	Elution stop collect	8	448	High	multivariate DoE

Step	Unit operation	Parameter	Sev.	Init. RPN	Prior-ity	Study type
7	Cation Exchange Chromatography	Elution flow rate	4	96	Low	univariate
8	Anion Exchange Chromatography	Load pH	10	700	High	multivariate DoE
8	Anion Exchange Chromatography	Equil/Wash-1 conductivity	6	336	Medium	multivariate DoE
8	Anion Exchange Chromatography	Load conductivity	10	700	High	multivariate DoE
8	Anion Exchange Chromatography	Protein load	6	336	Medium	multivariate DoE
8	Anion Exchange Chromatography	Operating flow rate	10	700	High	univariate (justified)
9	Small-Virus Retentive Filtration	Filtration volume (load)	10	700	High	multivariate DoE
9	Small-Virus Retentive Filtration	Filtration pressure	10	700	High	multivariate DoE
10	Ultrafiltration / Diafiltration	Number of diavolumes	4	96	Low	univariate
10	Ultrafiltration / Diafiltration	Transmembrane pressure	4	96	Low	univariate
10	Ultrafiltration / Diafiltration	Final DS concentration	4	96	Low	univariate

The assignment is prospective and it is revisited when the studies report. A factor that shows no effect is not removed from the record, but stays in the knowledge space as evidence that the attribute is robust to it over the range screened, and it is classified on that basis in the study report. A factor whose effect is larger than this assessment expected may require its design to be augmented, and that decision belongs to the study and not to this document. The assignment predicts no classification and no design space, and it is bounded by the parameter list in Table 5.2 and by the platform knowledge set out in §6.

6 Assumptions and limitations

The assessment rests on platform knowledge, and platform knowledge is an assumption until the studies test it. A-Mab is developed on the humanized IgG1 platform established by X-Mab, Y-Mab and Z-Mab and extended by A-Mab clinical and engineering experience, and the mechanistic expectations in §4, together with the ranges those expectations support, come from that lineage and from no other source. Where a platform expectation is wrong, this assessment will have placed the parameter in the wrong band, and the characterization study is what finds it.

The parameter list is assumed to be complete. Only parameters that are set or controlled by the manufacturing instructions are assessed here. Raw-material attributes, the equipment differences between the sending and the receiving site, and analytical method performance are assessed elsewhere, in PTP-001 for site equivalency and in the method validation records for the assays.

One attribute has no parameter ranked against it. Leached Protein A is set by the capture step, and no operating parameter of that step was linked to it, because the attribute is governed by resin age, sanitization and cycle number and not by an operating set-point. Resin life-cycle control (SOP-2008) is the control element for the attribute, the capture study measures it as a response (PCP-005), and the polishing steps clear it. The gap is in the parameter ranking and not in the control of the attribute.

Occurrence and detection are judgements at this point and not estimates from data, so the initial risk priority number is a ranking device on an ordinal scale, and differences between adjacent scores should not be read as ratios of one risk to another. Severity is the better grounded of the three axes, because it is inherited from an attribute criticality that was assessed against clinical and non-clinical evidence (CMC Biotech Working Group 2009).

No parameter scored below the study threshold, so the filter excluded nothing. The assessment is conservative in the direction of more study, which is defensible before data exist and would not be defensible afterwards. The scope will narrow as the studies report and the conservative defaults are replaced by measured effects.

The assessment assumes that a qualified scale-down model can represent each step. It does not assume that qualification has happened. Qualification against at-scale data is a deliverable of each plan, and a study whose scale-down model fails qualification does not deliver the range that this document assigned to it.

7 Outputs and downstream use

Each step's assignment is handed to one characterization plan, as listed in Table 7.1.

Table 7.1: The plan that consumes each step's assignment.

Step	Unit operation	Consuming plan	Parameters	Multivariate	Univariate
3	Production Bioreactor	PCP-003	9	5	4
4	Harvest and Clarification	PCP-004	3	0	3
5	Protein A Chromatography	PCP-005	6	4	2
6	Low-pH Viral Inactivation	PCP-006	4	3	1
7	Cation Exchange Chromatography	PCP-007	5	4	1
8	Anion Exchange Chromatography	PCP-008	5	4	1
9	Small-Virus Retentive Filtration	PCP-009	2	2	0
10	Ultrafiltration / Diafiltration	PCP-010	3	0	3

PCMP-001 schedules the plans and states the scale-down and statistical approach they share. Each report classifies the parameters of its own step against the evidence that its study produced, and PCMR-001 consolidates the classifications, the capability and the cumulative viral clearance for the whole process. This assessment is therefore revised whenever a study reports a result that changes a ranking, and the revision is recorded against the plan it affects.

8 Approvals

This assessment is approved by the functions recorded in Table 1. Approval confirms that the scope in Table 5.2 is the scope the characterization plans will execute, and that the assumptions in §6 are accepted.

References

CMC Biotech Working Group. 2009. *A-Mab: A Case Study in Bioprocess Development*. CASSS.

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International Council for Harmonisation. 2023b. *ICH Q9(R1): Quality Risk Management*. ICH.

U.S. Food and Drug Administration. 2011. *Guidance for Industry: Process Validation — General Principles and Practices*. FDA.